Quantum States

With a View Toward Homological Algebra

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Introduction

My work is about **states** of quantum spin systems with focus on:

- Structures underlying entire classes of states, including stabilizer states, invertible states, and KMS states
- Analysis using mathematical techniques, including homological algebra, operator theory, and complex analysis

Content

For the hour, I concentrate on stabilizer states and their *homological invariants*.

1. Translation-invariant stabilizer states
2. An algebraic bent
3. Homological tale
4. Higher category, Grothendieck duality and all that

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Quantum spin system

A *quantum spin system* consists of an infinite discrete space (Γ, d), often called a lattice, as well as an algebra of observables $\mathscr{A}$. Let $P_0(\Gamma)$ be the set of finite subsets of Γ. The algebra of *local observables* is

$$\mathscr{A}_{\mathrm{loc}} = \varinjlim_{X \in \mathcal{P}_0(\Gamma)} \bigotimes_{x \in X} M_{d_x}, \tag{1}$$

where M_{d_x} is the algebra of $d_x \times d_x$ matrices with $\sup_{x \in X} d_x < \infty$. We also denote $\bigotimes_{x \in X} M_{d_x}$ by $\mathscr{A}_X$ for all $X \in \mathcal{P}_0(\Gamma)$. The C^*-algebra $\mathscr{A}$ of *quasi-local observables* is the norm completion of $\mathscr{A}_{\mathrm{loc}}$.

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For us, $\Gamma = \mathbb{Z}^D$ for any positive integer D and d_x is constant. Then $\mathbb{Z}^D$ acts on $\mathscr{A}_{\mathrm{loc}}$ and $\mathscr{A}$ by translation.

Pure states

A *state* is a positive linear function $\psi : \mathscr{A} \to \mathbb{C}$ such that $\psi(\mathsf{Id}) = 1$. Here $\mathsf{Id} \in \mathscr{A}$ is the identity operator. The set $\mathfrak{S}$ of all states is clearly convex. A state is called *pure* if it is extremal[1] and *mixed* otherwise.

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Sometimes, we want to "stack" two systems $\mathscr{A}_1$ and $\mathscr{A}_2$ to produce a new system $\mathscr{A}_1 \otimes \mathscr{A}_2$. For two respective states ψ_1 and ψ_2, this produces a new state $\psi_1 \otimes \psi_2$ on the stacked system.

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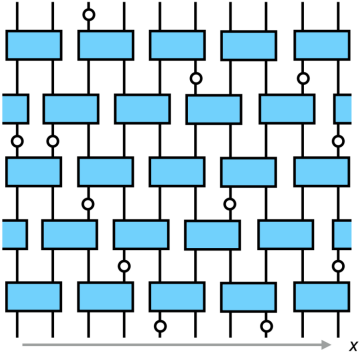
Unitary action and circuit

Let $\mathscr{U}(\mathscr{A}_{\mathrm{loc}})$ denote the group of unitary elements in $\mathscr{A}_{\mathrm{loc}}$. For any $\mathcal{U} \in \mathscr{U}(\mathscr{A}_{\mathrm{loc}})$ and $\psi \in \mathfrak{S}$, a new state $\psi_{\mathcal{U}}$ is defined by

$$\psi_{\mathcal{U}}(\mathcal{A}) = \psi(\mathcal{U}^{*}\mathcal{A}\mathcal{U}),$$

where $\mathcal{A} \in \mathscr{A}$. Thus it defines an action of $\mathscr{U}(\mathscr{A}_{\mathrm{loc}})$ on $\mathfrak{S}$.



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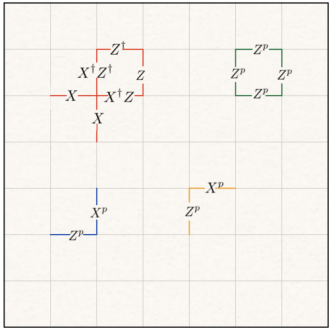
- Additionally, $\mathscr{L}$ is maximal abelian in $\mathscr{P}$.
- Finally, we assume the state is translation-invariant, i.e., $\mathscr{L}$ is closed under translation of the lattice.

Examples

Many interesting quantum phases are realized as stabilizer states, e.g., toric codes (and higher dimensional generalization), X-cube model, Haah's code etc.



$$\sigma_A = \begin{bmatrix} x^{-1}-1 & 0 & 0 & p \\ y^{-1}-1 & 0 & px & 0 \\ 1-y & p+py & p & 0 \\ x-1 & p+px & 0 & p\bar{y} \end{bmatrix}$$

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From stabilizer state to stabilizer module

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Therefore, we look at $\mathbb{Z}^D$ closed maximal abelian subgroups of $\mathscr{P}$.
Any $\mathcal{A}, \mathcal{B} \in \mathscr{P}$ satisfies the commutation relation

$$\mathcal{A}\mathcal{B} = \xi \mathcal{B}\mathcal{A},$$

where ξ is always a root of unity.

From stabilizer state to stabilizer module

We employ the following transformation:

- map $\mathscr{P}$ to a free R-module P, where $R = \mathbb{Z}_n[\mathbb{Z}^D] \cong \mathbb{Z}_n[x_1, x_1^{-1}, \cdots, x_D, x_D^{-1}]$.[2]

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- Let $L^\omega := \{x \in P \mid \omega(x, l) = 0 \text{ for all } l \in L\}$. Then $\mathscr{L}$ being abelian maximal translates to $L^\omega = L$, aka *Lagrangian*.

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In summary, all translation-invariant stabilizer states is in bijection with all Lagrangian submodules of a free module!

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[3] R is Gorenstein for any n.

“Functoriality”

What we have is much more than a set-theoretic bijection:

- Direct sum $L_1 \oplus L_2 \subset P_1 \oplus P_2$ corresponds to stacking.
- Quotient by ideal $(x_1^m - 1, \cdots, x_D^m - 1)$ puts the system on a size m torus.
- Elementary symplectic automorphisms $\mathscr{E}$ on P corresponds to Clifford circuits and translation.
- Symplectic automorphisms $\mathscr{S}$ corresponds to Clifford QCA's[4].
- Symplectic K_1 group $\mathscr{S}/\mathscr{E}$ equals the abelian group of non-trivial Clifford QCA's.
- and so on...

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[4] QCA's are (outer) automorphisms of $\mathscr{A}_{\text{loc}}$ preserving locality.

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Topological charges

To further understand and even classify stabilizer states, we define the following invariant:

$$Q^i = \mathrm{Ext}_R^{i+1}(\overline{P/L}, R), \qquad i \geq 0. \tag{2}$$

Theorem (Blazej Ruba, B.Y.)

Q^0 is isomorphic to the R-module of topological point excitations of the stabilizer state. Moreover, the excitations are mobile if and only if Q^0 has Krull dimension 0.

Corollary

Any two-dimensional stabilizer state has only mobile point excitations because $\dim(\mathrm{Ext}_R^{i+1}(\overline{P/L}, R)) \leq D - i - 1$.

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$$0 \to (P/L)^* \to P^* \to L^* \to \mathrm{Ext}^1_R(\bar{P}/\bar{L}, R) \to 0,$$

and

$$\mathrm{Ext}^{i+1}_R(\bar{P}/\bar{L}, R) \cong \mathrm{Ext}^i_R(\bar{L}, R),$$

for $i > 0$.

$M^* := \mathrm{Hom}_R(M, R)$. Thus $Q^0 \cong L^*/(P^*/(P/L)^*)$, that is all 'syndromes' $s : L \to R$ modulo those that extends to some $\tilde{s} : P \to R$.

Examples

Models	Q^0	Q^1	Q^2
3D $\mathbb{Z}_n$-toric code	$\mathbb{Z}_n$	$\mathbb{Z}_n$	0
4D $\mathbb{Z}_n$-toric code	$\mathbb{Z}_n$	0	$\mathbb{Z}_n$
4D $\mathbb{Z}_n$ 2-form toric code	0	$\mathbb{Z}_n \oplus \mathbb{Z}_n$	0
Haah's code	E	0	0
X-cube model	F	0	0

$$E = R/(1+x+y+xy, 1+y+z+yz, 1+z+x+xz) \oplus R^2/\mathrm{im}\,\tau,$$

with

$$\tau = \begin{pmatrix} 1+z & 0 \\ 1+x & 1+x \\ 0 & 1+y \end{pmatrix}.$$

$$F = R/(1+xy+yz+zx, 1+x+y+z) \oplus R/(1+\bar{x}+\bar{y}+\bar{z}, 1+\bar{x}\bar{y}+\bar{y}\bar{z}+\bar{z}\bar{x}).$$

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Notice $\dim E = 1 > 0$ and $\dim F = 1 > 0$. There exists states with $\dim Q^i > 0$ for $i > 0$.

Higher charges

Intuitively, Q^i should capture excitations that are dimension i membrane-like. However, a rigorous interpretation has not been found unless we restrict to (what we call) fully mobile cases with $\dim Q^i = 0$ for all i.

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Proposition (Blazej Ruba, B.Y.)

We have $Q^i \cong \check{\mathrm{H}}^{i+1}(\mathbf{t}, P/L)$ *for* $0 \leq i \leq D-2$, *where* $t = (x_1 - 1, \cdots, x_D - 1)$. $\check{\mathrm{H}}$ *is the derived functor of taking torsion submodule known as local cohomology.*

Local cohomology can be computed using a specific chain complex called the Čech complex whose cochains indeed represents dimension i membrane excitations.

Braiding as cup product

Definition

Let $\varphi \in \check{\mathrm{H}}^p(\mathbf{t}, P/L)$, $\psi \in \check{\mathrm{H}}^q(\mathbf{t}, P/L)$. *The cup product yields an element*

$$\overline{\varphi} \smile \delta\psi \in \check{\mathrm{H}}^{p+q}(\mathbf{t}, \overline{P/L} \otimes_R L), \tag{3}$$

where δ *is the connecting homomorphism* $\check{\mathrm{H}}^q(\mathbf{t}, P/L) \longrightarrow \check{\mathrm{H}}^{q+1}(\mathbf{t}, L)$ *in a long exact sequence. Using the map (with a slight abuse of notation) in Čech cohomology induced by the symplectic pairing* $\omega : \overline{P/L} \otimes_R L \longrightarrow R$ *we obtain a class*

$$\omega(\overline{\varphi} \smile \delta\psi) \in \check{\mathrm{H}}^{p+q}(\mathbf{t}, R). \tag{4}$$

This class is trivial unless $p + q = D$.

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Categorification?

Physically, (fully mobile) stabilizer states on $\mathbb{Z}^D$ should form a D-category. 1-morphisms are $(D-1)$-dimensional interfaces, and 2-morphisms are $(D-2)$-dimensional defects on a $(D-1)$-dimensional interface and so on...

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Can this be rigorously defined?

I believe the task is much more manageable on the purely algebraic setting of Lagrangian submodules.

Grothendieck duality

Grothendieck local duality

1 language

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From Wikipedia, the free encyclopedia

In commutative algebra, **Grothendieck local duality** is a duality theorem for cohomology of modules over local rings, analogous to Serre duality of coherent sheaves.

Statement [edit]

Suppose that R is a Cohen–Macaulay local ring of dimension d with maximal ideal m and residue field $k = R/m$. Let $E(k)$ be a Matlis module, an injective hull of k, and let $\overline{\Omega}$ be the completion of its dualizing module. Then for any R-module M there is an isomorphism of modules over the completion of R:

$$\mathrm{Ext}^i_R(M, \overline{\Omega}) \cong \mathrm{Hom}_R(H_m^{d-i}(M), E(k))$$

where H_m is a local cohomology group.

There is a generalization to Noetherian local rings that are not Cohen–Macaulay, that replaces the dualizing module with a dualizing complex.

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Is braiding Grothendieck duality?

Grothendieck duality is a non-degenerate pairing between Ext^i and $\check{\mathrm{H}}^{D-i}$. Meanwhile our braiding in terms of cup product is a pairing between $\check{\mathrm{H}}^i$ and $\check{\mathrm{H}}^{D-i}$ suspected to be non-degenerate. Since $\mathrm{Ext}^i \cong \check{\mathrm{H}}^i$ for fully mobile cases, are these two pairings one and the same?

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We simply don't know yet. If they were, it would be a beautiful phenomenon. More importantly, this would confirm that braiding is non-degenerate, which suggests fully mobile stabilizer states are abelian TQFT's.

Acknowledgement

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- Audience for their time and attention
- Anton, Nikita, and Blazej for keeping me honest

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Questions

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